

FRANCESCO BARILE

Bayesian Statistics · Network Modeling · Computational Statistics

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RESEARCH PROFILE

My research lies at the intersection of **Bayesian statistics**, **Bayesian nonparametrics**, **network science**, and **computational statistics**. I develop flexible statistical models and scalable computational methods for the analysis of complex, heterogeneous, and high-dimensional data.

A central theme of my work is the development of **flexible models for networked data**. I am also interested in modeling **dependence** across related populations and developing Bayesian nonparametric methods that accommodate heterogeneity across distributions while enabling efficient statistical inference.

My research takes an **application-oriented perspective**, with the goal of addressing practical challenges through methodological advances in statistical modeling and inference.

ACADEMIC POSITIONS

University of Milano-Bicocca, Milan, Italy

• **Postdoctoral Research Fellow**

Mar 2025 – present

(*Assegno di Ricerca*, pursuant to Article 22 of Italian Law No. 240/2010)

Department of Economics, Management and Statistics

Project: Joint Bayesian Learning of Related Clinical Studies [\[link\]](#)

Project PI: Prof. Bernardo Nipoti, University of Milano-Bicocca

Partner unit coordinator: Prof. Francesco C. Stingo, University of Florence

Funding: Fondazione Cariplo

EDUCATION

University of Milano-Bicocca, Milan, Italy

• **Ph.D. in Statistics**

2020 – 2024

Department of Economics, Management and Statistics

Defense: Feb. 14, 2025

Thesis: “Advances in Bayesian Methods: Nonparametric Network Modeling and Scalable Computations” **Final evaluation: Excellent**

Supervisor: Prof. Bernardo Nipoti

University of Calabria, Rende, Italy

• **M.Sc. in Finance and Insurance** (English)

2016 – 2019

Department of Economics, Statistics and Finance

Thesis: “Estimating Stochastic Models for Commodity Futures”

110/110 cum laude

Supervisor: Prof. Arturo Leccadito

• **B.Sc. in Statistics for Business and Insurance**

2012 – 2015

Department of Economics, Statistics and Finance

Thesis: “Test di Wilcoxon: applicazioni in ambito medico-farmaceutico”

110/110 cum laude

Supervisor: Prof. Agostino Tarsitano

RESEARCH INTERESTS

Bayesian statistics; Bayesian nonparametrics; network modeling; statistical learning; computational statistics.

PUBLICATIONS

REFEREED JOURNAL ARTICLES

- [1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025). Bayesian nonparametric modeling of heterogeneous populations of networks. *Bayesian Analysis*. Advance Publication. DOI: [10.1214/26-BA1588](https://doi.org/10.1214/26-BA1588). Code: [\[GitHub\]](#)

REFEREED CONFERENCE PROCEEDINGS

- [1] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2023). Flexible modelling of heterogeneous populations of networks: a Bayesian nonparametric approach. In C. Rampichini, M. La Rocca, P. Coretto, G. Giordano, M.L. Parrella (Eds.), *Book of Abstracts and Short Papers CLADAG 2023*, p. 69. Pearson. ISBN: 978-88-9193-563-2. [\[link\]](#)
- [2] **Barile, F.**, Lunagómez, S., & Nipoti, B. (2025). Clustering multiple network data with an application to the EU food trade market. In A. Pollice & P. Mariani (Eds.), *Methodological and Applied Statistics and Demography II: SIS 2024, Short Papers, Solicited Sessions*, pp. 68–72. Springer. ISBN: 978-3-031-64350-7. DOI: [10.1007/978-3-031-64350-7](https://doi.org/10.1007/978-3-031-64350-7).
- [3] **Barile, F.**, Dallari, S., Pacifici, C., & Grün, B. (2025). Assessing neuron response to external stimuli with a data-driven procedure for spike train extraction and GAMLSS regressions. In A. Canale, A. Luati, S. Mazzuco, R. Piccarreta, N. Sartori, P. Secchi (Eds.), *Advances in Neural Data Science*, pp. 35–55. Springer. ISBN: 978-3-031-70638-7. DOI: [10.1007/978-3-031-70638-7_3](https://doi.org/10.1007/978-3-031-70638-7_3).

MANUSCRIPTS UNDER REVIEW

- [1] **Barile, F.** & Nemeth, C. (2024). Control variate-based stochastic sampling from the probability simplex. *Manuscript under review*. Earlier version available at: *arXiv*: [\[2410.00845\]](https://arxiv.org/abs/2410.00845). Code: [\[GitHub\]](#)
- [2] **Barile, F.**, Capozio, S. & Nipoti, B. (2026). Efficient Bayesian inference for multiple network data. *Manuscript under review*. *arXiv*: [\[2610.01532\]](https://arxiv.org/abs/2610.01532). Code: [\[GitHub\]](#)

MANUSCRIPTS IN PREPARATION

- [1] **Barile, F.**, Borrotti, M., Nipoti, B., Poli, G. & Stingo, F. C. (2026+). Data Integration and Missing Imputation for Joint Bayesian Learning under Multiple Nonignorable Missing Mechanisms. *In preparation*.
- [2] **Barile, F.** & Nipoti, B. (2026+). Dependent Bernstein–Dirichlet priors for multiscale density estimation. *In preparation*.
- [3] **Barile, F.**, Nieto-Barajas, L. E. & Nipoti, B. (2026+). A Network of Dependent Dirichlet Processes. *In preparation*.
- [4] **Barile, F.**, Lunagómez, S. & Nemeth, C. (2026+). Covariate-dependent modeling of hypergraph data. *In preparation*.

PH.D. THESIS

- [1] **Barile, F.** (2025). *Advances in Bayesian Methods: Nonparametric Network Modeling and Scalable Computations*. Ph.D. thesis, XXXVI cycle. [\[link\]](#)

RESEARCH VISITS

- **Visiting Researcher**

Nov 2023 – Aug 2024

Lancaster University, Lancaster, United Kingdom
School of Mathematical Sciences
Supervisor: Prof. Christopher Nemeth

- **Visiting Graduate Student (Research Only)**

Aug – Dec 2018

University of Western Ontario, London, Canada
Department of Statistical and Actuarial Sciences
Supervisor: Prof. Rogemar Mamon
Research topic: Statistical modeling of lumber price dynamics

RESEARCH SOFTWARE

DPM_CER Bayesian nonparametric modeling of heterogeneous populations of networks. [[GitHub](#)]
BayesCER Bayesian inference for multiple network data. [[GitHub](#)]
CV_SCIR Control-variate-based stochastic sampling on the probability simplex. [[GitHub](#)]
sBNPi Bayesian nonparametric imputation for structural missing data. [[GitHub](#)]

TEACHING

■ University of Milano-Bicocca, Department of Economics, Management and Statistics

- **Inferential Statistics / Advanced Statistics** [English] (joint class) 2026/2027

Master's Degrees in International Economics and
Marketing and Global Markets

Adjunct Lecturer

- **Industrial Economics and Dynamics** [Italian] 2021/2022

Master's Degree in Statistical and Economic Sciences

Teaching Assistant

- **Inferential Statistics** [English], Master's Degree in International Economics 2025/2026
- **Inferential Statistics** [English], Master's Degree in International Economics 2024/2025
- **Advanced Statistics** [English], Master's Degree in Marketing and Global Markets 2024/2025
- **Inferential Statistics** [English], Master's Degree in International Economics 2023/2024
- **Advanced Statistics** [English], Master's Degree in Marketing and Global Markets 2023/2024
- **Inferential Statistics** [English], Master's Degree in International Economics 2022/2023

Teaching experience includes statistical inference, probability, descriptive statistics, sampling distributions, estimation, hypothesis testing, correlation, and quantitative data analysis.

PRESENTATIONS AND SEMINARS

INVITED PRESENTATIONS

- CFE-CMStatistics 2026, HTW Berlin, University of Applied Sciences, Berlin, Germany Dec 12–14, 2026
- StaTalk 2025, University of Milano-Bicocca, Milan, Italy Jun 13–14, 2025

• ECOSTA 2023, Waseda University, Tokyo, Japan Aug 1–3, 2023

CONTRIBUTED PRESENTATIONS

• BAYSM 2023, Online Nov 13–17, 2023

POSTERS

• ISBA 2026, Nagoya, Japan Jun 28 – Jul 3, 2026
• BayesComp 2025, Singapore Jun 16–20, 2025
• SISBayes 2025, Padova, Italy Sep 4–5, 2025
• Complex Graphical Models for Biological Network Science, Padova, Italy Sep 4, 2024
• ISBA 2024, Venice, Italy Jul 1–7, 2024
• IMS ICSDS 2022, Florence, Italy Dec 13–16, 2022

SEMINARS

• Computational Statistics and Machine Learning Seminar Series, School of Mathematical Sciences, Lancaster University, Lancaster, UK Feb 2024

WORKSHOPS

• Data Research Camp 2022, VIU, San Servolo Island, Venice, Italy Jul 12–15, 2022

ACADEMIC SERVICE

CONFERENCE ORGANIZATION

• Organizer and Chair, “Recent advances in Bayesian variable selection and shrinkage”, Dec 2026
CFE-CMStatistics 2026 [[link](#)]

PEER REVIEW

• Reviewer for *Computational Statistics & Data Analysis*
• Reviewer for *Communications in Statistics – Simulation and Computation*
• Reviewer for *IEEE Journal of Biomedical and Health Informatics*

PROFESSIONAL MEMBERSHIPS

• International Society for Bayesian Analysis (ISBA), Junior ISBA, BNP-ISBA
• Italian Statistical Society (SIS), Young SIS

STUDENT SUPERVISION

• Co-supervisor of one Master’s thesis, Master’s Degree in Statistical and Economic Sciences, University of Milano-Bicocca

SCHOLARSHIPS AND AWARDS

• **ISBA 2026 Junior Travel Support** 2026
International Society for Bayesian Analysis
• **BayesComp 2025 Junior Travel Support** 2025
Bayesian Computation Conference, Singapore
• **Extra-UE Mobility Grant** 2023
University of Milano-Bicocca
• **Ph.D. Scholarship** 2020
University of Milano-Bicocca
• **Overseas Research Scholarship** 2018
University of Calabria

COMPUTATIONAL SKILLS

Programming	R, C++, Python (basic), MATLAB (basic), SQL, Java, L ^A T _E X
Statistical software	SAS, Stata, RStudio
Research computing	C++-based R package development; parallel computing; Bayesian computation; Monte Carlo methods; reproducible statistical research; version control with Git/GitHub; scientific programming.

ADDITIONAL PROFESSIONAL EXPERIENCE

Junior Data Scientist , CRIF S.p.A., Cosenza, Italy	Apr 2019 – Sep 2020
Degree Program Tutor , University of Calabria, Rende, Italy	Feb – Jul 2018
	Apr – Oct 2017

LANGUAGES

Italian	Mother tongue
English	Fluent
French	Basic

Le dichiarazioni rese nel presente curriculum sono da ritenersi rilasciate ai sensi degli artt. 46 e 47 del D.P.R. 445/2000. Autorizzo la pubblicazione ai sensi del D.Lgs. n. 33/2013 e il trattamento dei dati personali ai sensi del Regolamento (UE) 2016/679 e del D.Lgs. n. 196/2003.

Milan, October 9, 2026